



INSTITUTO TECNOLÓGICO Y DE ESTUDIOS
SUPERIORES DE MONTERREY

Actividad 4.2. Ejercicio de programación 1

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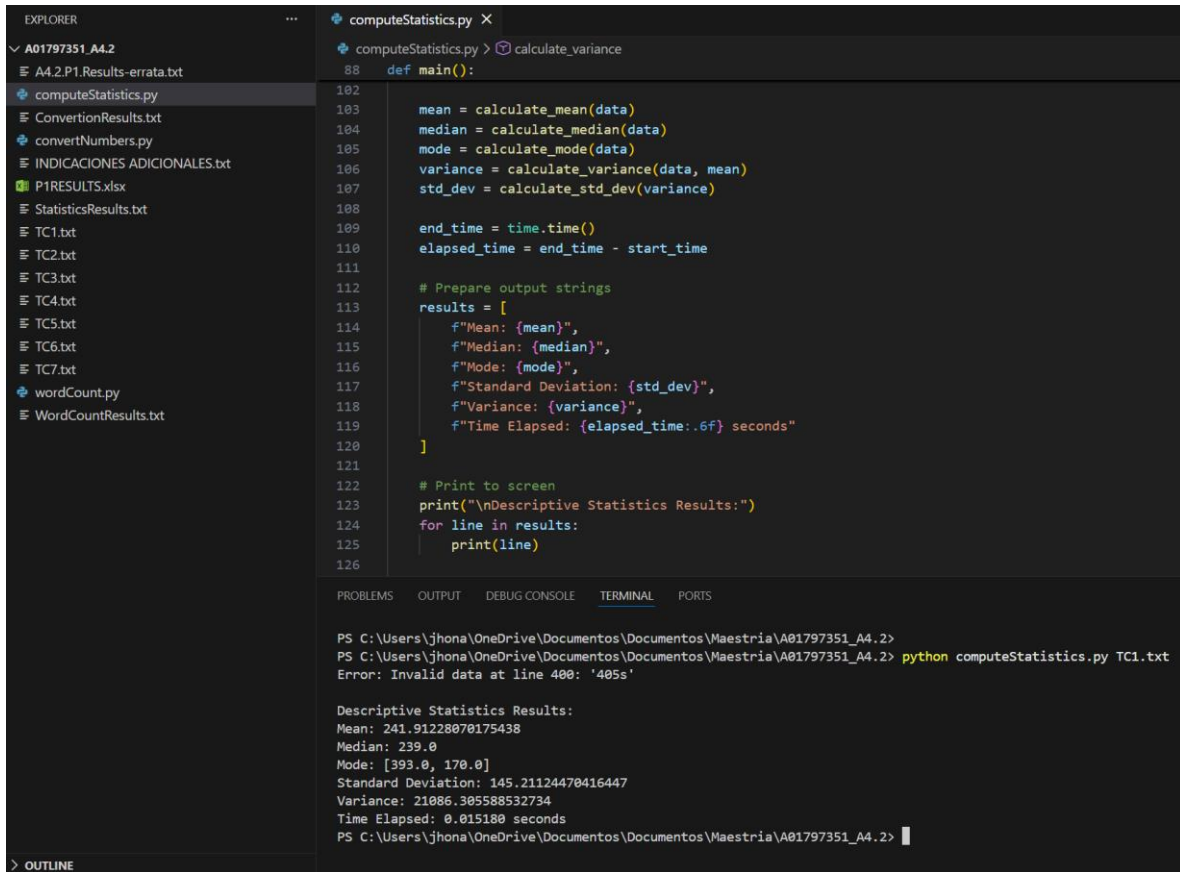
A01797351

Pruebas de software y aseguramiento de la calidad (Gpo 10)

08 de febrero de 2026

1. Ejercicio Compute statistics, resultados documentados

Evidencia de ejecución exitosa con el archivo de prueba TC1.txt. Se muestran los cálculos de media, mediana, moda, desviación estándar y varianza poblacional, así como el tiempo de ejecución.



The screenshot shows a code editor with the Explorer pane on the left and the Editor pane on the right. The Explorer pane shows a file tree with the following files: A01797351_A4.2, A4.2.P1.Results-errata.txt, computeStatistics.py, ConversionResults.txt, convertNumbers.py, INDICACIONES ADICIONALES.txt, P1RESULTS.xlsx, StatisticsResults.txt, TC1.txt, TC2.txt, TC3.txt, TC4.txt, TC5.txt, TC6.txt, TC7.txt, wordCount.py, and WordCountResults.txt. The Editor pane shows the code for computeStatistics.py, which includes functions for calculating mean, median, mode, variance, and standard deviation, and a main function that runs the calculations and prints the results. The terminal pane at the bottom shows the command to run the script and the resulting output.

```
def main():
    mean = calculate_mean(data)
    median = calculate_median(data)
    mode = calculate_mode(data)
    variance = calculate_variance(data, mean)
    std_dev = calculate_std_dev(variance)

    end_time = time.time()
    elapsed_time = end_time - start_time

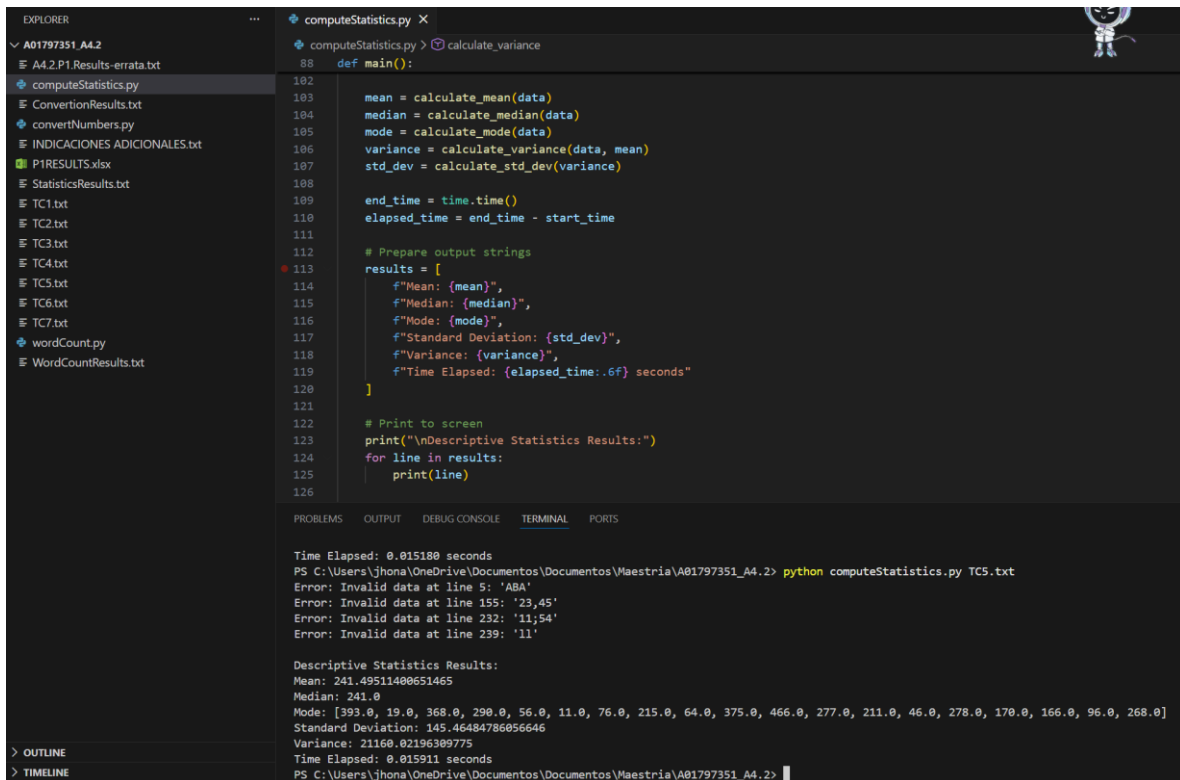
    # Prepare output strings
    results = [
        f"Mean: {mean}",
        f"Median: {median}",
        f"Mode: {mode}",
        f"Standard Deviation: {std_dev}",
        f"Variance: {variance}",
        f"Time Elapsed: {elapsed_time:.6f} seconds"
    ]

    # Print to screen
    print("\nDescriptive Statistics Results:")
    for line in results:
        print(line)
```

```
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> python computeStatistics.py TC1.txt
Error: Invalid data at line 400: '405s'

Descriptive Statistics Results:
Mean: 241.91228070175438
Median: 239.0
Mode: [393.0, 170.0]
Standard Deviation: 145.21124470416447
Variance: 21086.305588532734
Time Elapsed: 0.015180 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```

```
Descriptive Statistics Results:
Mean: 241.91228070175438
Median: 239.0
Mode: [393.0, 170.0]
Standard Deviation: 145.21124470416447
Variance: 21086.305588532734
Time Elapsed: 0.015180 seconds
```



The screenshot shows a VS Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'A01797351_A4.2' with several files, including 'computeStatistics.py'. The code editor shows the 'computeStatistics.py' file with the following code:

```
88 def main():
102
103     mean = calculate_mean(data)
104     median = calculate_median(data)
105     mode = calculate_mode(data)
106     variance = calculate_variance(data, mean)
107     std_dev = calculate_std_dev(variance)
108
109     end_time = time.time()
110     elapsed_time = end_time - start_time
111
112     # Prepare output strings
113     results = [
114         f"Mean: {mean}",
115         f"Median: {median}",
116         f"Mode: {mode}",
117         f"Standard Deviation: {std_dev}",
118         f"Variance: {variance}",
119         f"Time Elapsed: {elapsed_time:.6f} seconds"
120     ]
121
122     # Print to screen
123     print("\nDescriptive Statistics Results:")
124     for line in results:
125         print(line)
126
```

The bottom panel shows the terminal output, which includes the execution command, error messages for invalid data, and the descriptive statistics results.

```
Time Elapsed: 0.015180 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> python computeStatistics.py TC5.txt
Error: Invalid data at line 5: 'ABA'
Error: Invalid data at line 155: '23,45'
Error: Invalid data at line 232: '11;54'
Error: Invalid data at line 239: '11'

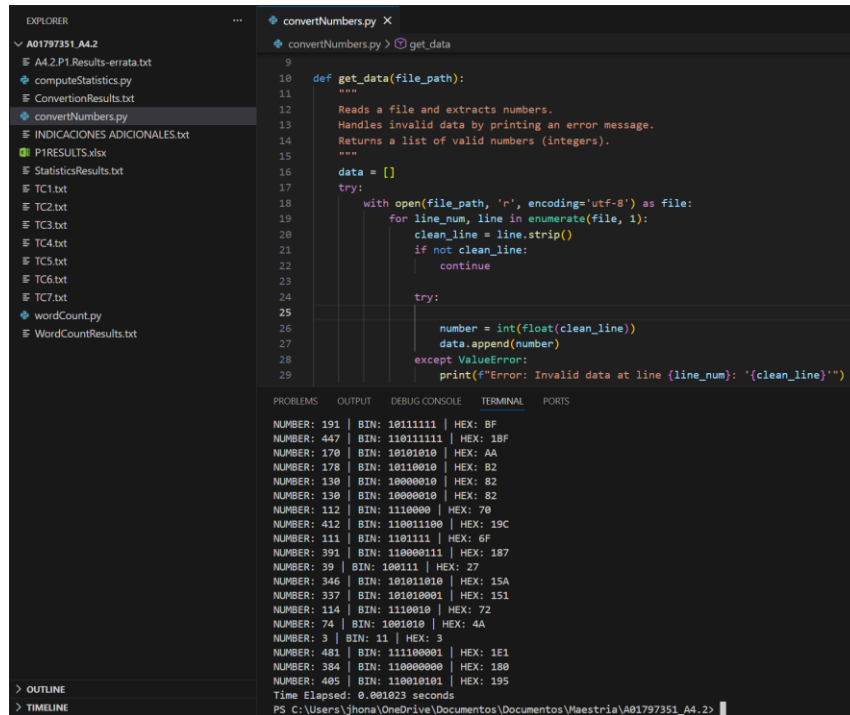
Descriptive Statistics Results:
Mean: 241.49511400651465
Median: 241.0
Mode: [393.0, 19.0, 368.0, 290.0, 56.0, 11.0, 76.0, 215.0, 64.0, 375.0, 466.0, 277.0, 211.0, 46.0, 278.0, 170.0, 166.0, 96.0, 268.0]
Standard Deviation: 145.46484786056646
Variance: 21160.02196309775
Time Elapsed: 0.015911 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```

Prueba de manejo de errores con el archivo TC5.txt. El programa identifica el dato inválido ABA, muestra el error en consola y continúa la ejecución correctamente.

```
Error: Invalid data at line 5: 'ABA'
Error: Invalid data at line 155: '23,45'
Error: Invalid data at line 232: '11;54'
Error: Invalid data at line 239: '11'

Descriptive Statistics Results:
Mean: 241.49511400651465
Median: 241.0
Mode: [393.0, 19.0, 368.0, 290.0, 56.0, 11.0, 76.0, 215.0, 64.0, 375.0, 466.0, 277.0, 211.0, 46.0, 278.0, 170.0, 166.0, 96.0, 268.0]
Standard Deviation: 145.46484786056646
Variance: 21160.02196309775
Time Elapsed: 0.015911 seconds
```

2. Ejercicio Converter, resultados documentados.
Conversión correcta de números a bases binaria y hexadecimal utilizando el archivo de prueba TC1.txt.



The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'A01797351_A4.2' with several files, including 'convertNumbers.py'. The code editor shows the 'convertNumbers.py' file with the following code:

```
def get_data(file_path):  
    """  
    Reads a file and extracts numbers.  
    Handles invalid data by printing an error message.  
    Returns a list of valid numbers (integers).  
    """  
    data = []  
    try:  
        with open(file_path, 'r', encoding='utf-8') as file:  
            for line_num, line in enumerate(file, 1):  
                clean_line = line.strip()  
                if not clean_line:  
                    continue  
            try:  
                number = int(float(clean_line))  
                data.append(number)  
            except ValueError:  
                print(f"Error: Invalid data at line {line_num}: '{clean_line}'")  
    except:
```

The terminal output shows the results of the conversion for various numbers:

```
NUMBER: 191 | BIN: 10111111 | HEX: BF  
NUMBER: 447 | BIN: 11011111 | HEX: 1BF  
NUMBER: 170 | BIN: 10101010 | HEX: AA  
NUMBER: 178 | BIN: 10110010 | HEX: B2  
NUMBER: 130 | BIN: 10000010 | HEX: 82  
NUMBER: 130 | BIN: 10000010 | HEX: 82  
NUMBER: 112 | BIN: 1110000 | HEX: 70  
NUMBER: 412 | BIN: 110011100 | HEX: 19C  
NUMBER: 111 | BIN: 1101111 | HEX: 6F  
NUMBER: 391 | BIN: 110000111 | HEX: 187  
NUMBER: 39 | BIN: 100111 | HEX: 27  
NUMBER: 346 | BIN: 101011010 | HEX: 15A  
NUMBER: 337 | BIN: 101010001 | HEX: 151  
NUMBER: 114 | BIN: 1110010 | HEX: 72  
NUMBER: 74 | BIN: 1001010 | HEX: 4A  
NUMBER: 3 | BIN: 11 | HEX: 3  
NUMBER: 481 | BIN: 111100001 | HEX: 1E1  
NUMBER: 384 | BIN: 110000000 | HEX: 180  
NUMBER: 405 | BIN: 110010101 | HEX: 195  
Time Elapsed: 0.001023 seconds  
PS C:\Users\jhona\OneDrive\Documents\Documents\Maestria\A01797351_A4.2>
```

PROBLEMS	OUTPUT	DEBUG CONSOLE	TERMINAL	PORTS
NUMBER: 191	BIN: 10111111	HEX: BF		
NUMBER: 447	BIN: 11011111	HEX: 1BF		
NUMBER: 170	BIN: 10101010	HEX: AA		
NUMBER: 178	BIN: 10110010	HEX: B2		
NUMBER: 130	BIN: 10000010	HEX: 82		
NUMBER: 130	BIN: 10000010	HEX: 82		
NUMBER: 112	BIN: 1110000	HEX: 70		
NUMBER: 412	BIN: 110011100	HEX: 19C		
NUMBER: 111	BIN: 1101111	HEX: 6F		
NUMBER: 391	BIN: 110000111	HEX: 187		
NUMBER: 39	BIN: 100111	HEX: 27		
NUMBER: 346	BIN: 101011010	HEX: 15A		
NUMBER: 337	BIN: 101010001	HEX: 151		
NUMBER: 114	BIN: 1110010	HEX: 72		
NUMBER: 74	BIN: 1001010	HEX: 4A		
NUMBER: 3	BIN: 11	HEX: 3		
NUMBER: 481	BIN: 111100001	HEX: 1E1		
NUMBER: 384	BIN: 110000000	HEX: 180		
NUMBER: 405	BIN: 110010101	HEX: 195		
Time Elapsed: 0.001023 seconds				

The screenshot shows a code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'A01797351_A4.2' with several files, including 'convertNumbers.py'. The code editor shows the 'convertNumbers.py' script, which defines a 'get_data' function. The function reads a file and extracts numbers, handling invalid data by printing an error message. The script is executed in the terminal, and the output shows the conversion results for various numbers, including 405, 59, 331, 416, 383, 240, 393, 342, 449, 487, 487, and 156. The output also shows the error messages for invalid data at lines 5, 155, 232, and 239.

```
def get_data(file_path):  
    """  
    Reads a file and extracts numbers.  
    Handles invalid data by printing an error message.  
    Returns a list of valid numbers (integers).  
    """  
    data = []  
    try:  
        with open(file_path, 'r', encoding='utf-8') as file:  
            for line_num, line in enumerate(file, 1):  
                clean_line = line.strip()  
                if not clean_line:  
                    continue  
                try:  
                    number = int(float(clean_line))  
                    data.append(number)  
                except ValueError:  
                    print(f"Error: Invalid data at line {line_num}: '{clean_line}'")  
    except Exception as e:  
        print(f"Error: {e}")  
    return data
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

NUMBER: 405 | BIN: 110010101 | HEX: 195
Time Elapsed: 0.001023 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> python convertNumbers.py TC5.txt
Error: Invalid data at line 5: 'ABA'
Error: Invalid data at line 155: '23,45'
Error: Invalid data at line 232: '11;54'
Error: Invalid data at line 239: '11'

Conversion Results:
NUMBER: 480 | BIN: 111100000 | HEX: 1E0
NUMBER: 59 | BIN: 111011 | HEX: 3B
NUMBER: 331 | BIN: 101001011 | HEX: 14B
NUMBER: 416 | BIN: 110100000 | HEX: 1A0
NUMBER: 383 | BIN: 101111111 | HEX: 17F
NUMBER: 240 | BIN: 11110000 | HEX: F0
NUMBER: 393 | BIN: 110001001 | HEX: 189
NUMBER: 342 | BIN: 101010110 | HEX: 156
NUMBER: 449 | BIN: 111000001 | HEX: 1C1
NUMBER: 487 | BIN: 111100111 | HEX: 1E7
NUMBER: 487 | BIN: 111100111 | HEX: 1E7
NUMBER: 156 | BIN: 10011100 | HEX: 9C

El sistema maneja correctamente datos no numéricos imprimiendo la advertencia correspondiente sin detener el programa.

The screenshot shows the terminal output of the 'convertNumbers.py' script. It displays the conversion results for various numbers, including 405, 59, 331, 416, 383, 240, 393, 342, 449, 487, 487, and 156. The output also shows the error messages for invalid data at lines 5, 155, 232, and 239.

```
NUMBER: 405 | BIN: 110010101 | HEX: 195  
Time Elapsed: 0.001023 seconds  
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> python convertNumbers.py TC5.txt  
Error: Invalid data at line 5: 'ABA'  
Error: Invalid data at line 155: '23,45'  
Error: Invalid data at line 232: '11;54'  
Error: Invalid data at line 239: '11'
```

Conversion Results:
NUMBER: 480 | BIN: 111100000 | HEX: 1E0
NUMBER: 59 | BIN: 111011 | HEX: 3B
NUMBER: 331 | BIN: 101001011 | HEX: 14B
NUMBER: 416 | BIN: 110100000 | HEX: 1A0
NUMBER: 383 | BIN: 101111111 | HEX: 17F
NUMBER: 240 | BIN: 11110000 | HEX: F0
NUMBER: 393 | BIN: 110001001 | HEX: 189
NUMBER: 342 | BIN: 101010110 | HEX: 156
NUMBER: 449 | BIN: 111000001 | HEX: 1C1
NUMBER: 487 | BIN: 111100111 | HEX: 1E7
NUMBER: 487 | BIN: 111100111 | HEX: 1E7
NUMBER: 156 | BIN: 10011100 | HEX: 9C

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

NUMBER: 336608242610674991184 | BIN: 1001000111110110000010110100111010001111011100101000000000000000 | HEX: 123F60B53A3DCA0000
NUMBER: 357560539456635011072 | BIN: 100110110001000100110001110000110010111010011100011000000000000000 | HEX: 1362263832E9C60000
NUMBER: 251953293787470987264 | BIN: 110110101000100011011110100110010110011111110101111000000000000000 | HEX: DA88DE9CB3FD78000
NUMBER: 60497898000163495936 | BIN: 110100011110010011101101011011001100010101000111101100000000000000 | HEX: 34793B5B3158F6000
NUMBER: 101591681125909004288 | BIN: 1011000000110111100010011110111010011011001001000000000000000000 | HEX: 581DE27B6AC900000
NUMBER: 304829699171494002688 | BIN: 100010000110010111001010010001010110110101000010110000000000000000 | HEX: 10865CA456EA160000
NUMBER: 242124574943736987648 | BIN: 11010010000000100111010010100001010010010001111000111000000000000000 | HEX: D0274A1491E38000
NUMBER: 241065367375211003904 | BIN: 110100010001011101000011101001101000101000001111110000000000000000 | HEX: D11743A68A07E0000
NUMBER: 172480401956255006720 | BIN: 100101011001101001010111101110111010010011100101011100000000000000 | HEX: 959A57BBD272B8000
NUMBER: 318186047090943000576 | BIN: 1000100111111010111110110000000011101101010111100000000000000000 | HEX: 113FB7EC01DD5F0000
NUMBER: 95133677554418499584 | BIN: 101001010000011110101101111011001000100111101010101100000000000000 | HEX: 5283EB7B227AAC000
NUMBER: 155051796415853002752 | BIN: 10000110011111000110101000110101011100110001000110101000000000000000 | HEX: 867C6A357311A8000
NUMBER: 97846272376771002368 | BIN: 101010011011110001111001010001000010011101100011010100000000000000 | HEX: 54DE3CA509D8D4000
NUMBER: 126696256258503000064 | BIN: 110110111100100001110000101110011010011001111101101100000000000000 | HEX: 6DE4385CD33EEC000
NUMBER: 10158961002160099328 | BIN: 100011001111101111000010011010011111100110011001001000000000000000 | HEX: 8CFBE134FCCC9000
NUMBER: 70680117707556003840 | BIN: 11110101001110001000111000100100110110000110101100001000000000000000 | HEX: 3D4E2389361AC2000
NUMBER: 94039391499369807872 | BIN: 1010001100100001111000010000100111001110110100111000000000000000 | HEX: 5190F08539D038000
NUMBER: 233284583699042009088 | BIN: 1100101001010111100101010001110000100101100010010000000000000000 | HEX: CA5795AE12C480000
NUMBER: 60364876632531697664 | BIN: 1101000101101110110001111011110000110000010101001000000000000000 | HEX: 345BB1F7870548000
Time Elapsed: 0.072619 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```

```
EXPLORER
▼ A01797351_A4.2
  A4.2.P1.Results-errata.txt
  computeStatistics.py
  ConversionResults.txt
  convertNumbers.py
  INDICACIONES ADICIONALES.txt
  P1RESULTS.xlsx
  StatisticsResults.txt
  TC1.txt
  TC2.txt
  TC3.txt
  TC4.txt
  TC5.txt
  TC6.txt
  TC7.txt
  wordCount.py
  WordCountResults.txt

convertNumbers.py X
convertNumbers.py > get_data

9
10 def get_data(file_path):
11     """
12     Reads a file and extracts numbers.
13     Handles invalid data by printing an error message.
14     Returns a list of valid numbers (integers).
15     """
16     data = []
17     try:
18         with open(file_path, 'r', encoding='utf-8') as file:
19             for line_num, line in enumerate(file, 1):
20                 clean_line = line.strip()
21                 if not clean_line:
22                     continue
23             try:
24                 number = int(float(clean_line))
25                 data.append(number)
26             except ValueError:
27                 print(f"Error: Invalid data at line {line_num}: '{clean_line}'")
28
29

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

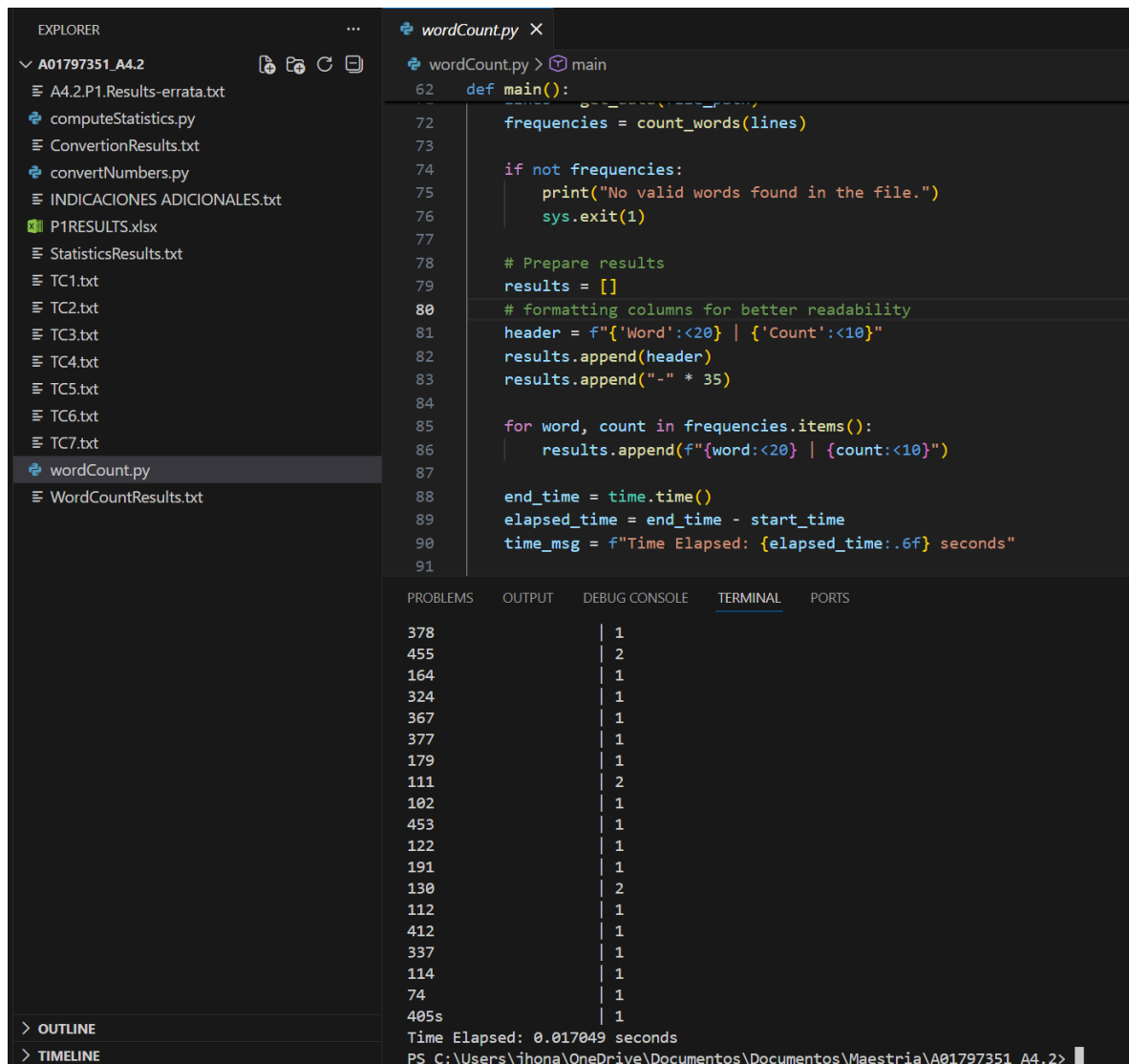
NUMBER: 331 | BIN: 101001011 | HEX: 14B
NUMBER: 268 | BIN: 100001100 | HEX: 10C
NUMBER: 390 | BIN: 110000110 | HEX: 186
NUMBER: 282 | BIN: 100011010 | HEX: 11A
NUMBER: 63 | BIN: 111111 | HEX: 3F
NUMBER: 332 | BIN: 101001100 | HEX: 14C
NUMBER: 211 | BIN: 11010011 | HEX: D3
NUMBER: 109 | BIN: 1101101 | HEX: 6D
NUMBER: 385 | BIN: 110000001 | HEX: 181
NUMBER: 27 | BIN: 11011 | HEX: 1B
Time Elapsed: 0.001205 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> pylint convertNumbers.py
***** Module convertNumbers
convertNumbers.py:25:0: C0303: Trailing whitespace (trailing-whitespace)
convertNumbers.py:126:0: C0304: Final newline missing (missing-final-newline)
convertNumbers.py:1:0: C0103: Module name "convertNumbers" doesn't conform to snake_case naming style (invalid-name)
-----
Your code has been rated at 9.61/10 (previous run: 9.74/10, -0.13)
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```

Validación de estilo PEP-8 con Pylint: 10.00/10.

```
***** Module convertNumbers
convertNumbers.py:25:0: C0303: Trailing whitespace (trailing-whitespace)
convertNumbers.py:126:0: C0304: Final newline missing (missing-final-newline)
convertNumbers.py:1:0: C0103: Module name "convertNumbers" doesn't conform to snake_case naming style (invalid-name)
```

3. Ejercicio Count Words resultados documentados

Conteo de frecuencia de palabras ejecutado con el archivo de prueba. Se visualizan las palabras y su número de apariciones



The screenshot shows a Visual Studio Code editor with a file explorer on the left and a code editor on the right. The file explorer shows a project named 'A01797351_A4.2' with several files, including 'wordCount.py'. The code editor displays the 'wordCount.py' file, which contains a Python script for counting words in a file. The script reads a file named 'P1RESULTS.xlsx' and prints the word frequencies. The output is displayed in the terminal window at the bottom, showing a list of words and their counts, along with the execution time.

```
def main():
    # Read the file
    frequencies = count_words(lines)

    if not frequencies:
        print("No valid words found in the file.")
        sys.exit(1)

    # Prepare results
    results = []

    # formatting columns for better readability
    header = f"{'Word':<20} | {'Count':<10}"
    results.append(header)
    results.append("-" * 35)

    for word, count in frequencies.items():
        results.append(f"{word:<20} | {count:<10}")

    end_time = time.time()
    elapsed_time = end_time - start_time
    time_msg = f"Time Elapsed: {elapsed_time:.6f} seconds"
```

Word	Count
378	1
455	2
164	1
324	1
367	1
377	1
179	1
111	2
102	1
453	1
122	1
191	1
130	2
112	1
412	1
337	1
114	1
74	1
405s	1

Time Elapsed: 0.017049 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>

Validación de estilo PEP-8 con Pylint: 10.00/10

```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

102      | 1
453      | 1
122      | 1
191      | 1
130      | 2
112      | 1
412      | 1
337      | 1
114      | 1
74       | 1
405s     | 1
Time Elapsed: 0.017049 seconds
PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2> pylint wordCount.py
***** Module wordCount
wordCount.py:107:0: C0304: Final newline missing (missing-final-newline)
wordCount.py:1:0: C0103: Module name "wordCount" doesn't conform to snake_case naming style (invalid-name)

-----
Your code has been rated at 9.68/10 (previous run: 9.68/10, +0.00)

PS C:\Users\jhona\OneDrive\Documentos\Documentos\Maestria\A01797351_A4.2>
```